DUPLICATE
ORIGINAL

Release Purchase Order for Master Research Agreement (No. 4600000582)
Release Purchase Order (RPO) No.21

Upon execution by the parties below, the Research Project specified herein shall be awarded and performed in accordance with the Master Research Agreement ("Agreement", No. 4600000582), which shall be incorporated herein in its entirety. The project specifications shall include the following:

a. The Research Project title:
Assessment of Iron and Steel Technologies from A Global Perspective and Study of China's Copper Supply Chain Towards Carbon Neutrality

b. A description of the purpose of the project:

This project has two primary objectives:
(1) To develop an optimization analysis framework for hydrogen-based steelmaking technologies that incorporates both regional characteristics and global implications. The framework aims to support decarbonization and low-carbon technology deployment in major steel-producing regions worldwide, while also providing insights for assessing emerging patterns in global steel trade.
(2) To investigate China's copper supply chain—focusing on copper resource flows, material metabolism, and medium- to long-term demand evolution—in order to better understand the supply-demand dynamics and potential risks associated with this critical mineral in the energy transition.

c. A detailed scope of work and implementation plan to include the contribution of both University and AAJ, as applicable, and all milestones and deliverable for each phase of the project (Attachment):
Scope of Work is shown in Attachment I of RPO No.21.

d. Duration of the project: Start Date: July 21, 2025, End Date: July 20, 2027.

e. A list of any funding requirements for this Research Project (Attachment):
None.

f. A proposal for who should act as Principal Investigator in the project:
Tsinghua University Principal Investigator of this CONTRACT shall be:
Contact Name: Prof. Zheng Li

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Telephone: +86 010-62795736
E-Mail: lz-dte@tsinghua.edu.cn

Co-PI : Dr. Tianduo Peng
Telephone : +86 010-62780968 +86 13671058362
E-Mail : ptd@mail.tsinghua.edu.cn

g. The Saudi Aramco project representative:
Saudi Aramco Representative for the purposes of administering this CONTRACT shall be:
Contact Name: Daniel De Castro Gomez
Telephone: +966 13 8808457
E-Mail: daniel.decastrogomez@aramco.com

Or another duly appointed delegate.
Saudi Aramco Representative must be copied on all reports and electronic communication between Tsinghua University and SAUDI ARAMCO.

h. A brief description of any confidential information that will be shared:
The confidential information is defined in article 8 of the Agreement.

i. A description of any background IP that will be used as part of the proposed project:
None.

j. A description of any pre-existing work that is being done by University in the proposed area of research, to include informing AAJ of any third Parties that are or that may become involved in the project:
None

k. Intellectual property ownership provisions, if different from the terms specified in this Agreement.
The Intellectual Property provisions remain the same as in the Master Research Agreement.

l. Financial Accounting:
Amount of Funding from AAJ:
 i. Amount for this action $400,000.00

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m. Invoicing Frequency:
i. ☐ Quarterly
ii. ☒ Other (refer to Attachment 1 of RPO No. 21)

n. Technical Reporting Requirements:
i. ☒ Progress Reports;
ii. ☐ Quarterly Meetings
iii. ☒ Final Report (90 days post term)

o. Other terms and conditions:
COMPANY shall acknowledge receipt of each of the above-mentioned deliverables.

The duly authorized representative of the University hereby agrees to accept the terms of this RPO as indicated by their signatures below.

Tsinghua University	Aramco Asia Japan K.K.
吴华强 李政	
Signature: ____________________	Signature: WALEED MURAD
Acknowledged and accepted by Principal Investigator:	Name: Waleed Murad
	Title: Representative Director

ENDORSED FOR SIGNATURE
20
DATE: 07/03/2025
BY:
ARAMCO ASIA
CONTRACTING UNIT

Contract Release Purchase Order (CRPO) No. 21

Scope of Work

Assessment of Iron and Steel Technologies from A Global Perspective and Study of China's Copper Supply Chain Towards Carbon Neutrality

1. INTRODUCTION

This document establishes the scope and means of initiating WORK to be performed by the Parties and describes or references the specifications, instructions, standards, and other documents, which Parties shall satisfy or adhere to in the performance of WORK.

The objective of this proposal is to develop a research project "Assessment of Iron and Steel Technologies from A Global Perspective and Study of China's Copper Supply Chain Towards Carbon Neutrality". The following sections are to be used as a plan to guide this research project.

2. BACKGROUND INFORMATION

Iron and steel industry is one of the core industries underpinning modern infrastructure and economic development. However, steel production comes at a significant environmental cost—it is the second-largest source of carbon emissions in China. Emission reductions in this sector are therefore crucial to China's low-carbon transition. Phase I of the project focused on the flexibility of hydrogen-based iron- and steelmaking systems powered by variable renewable energy and evaluated how this technological transformation could reshape the spatial distribution of China's steel industry.

Building on the outcomes of Phase I, we recognize that hydrogen-based iron- and steelmaking not only holds great promise in China, but also presents significant opportunities—and challenges—in major iron and steel-producing regions globally. Accordingly, Phase II aims to extend the flexibility and optimization analysis to key international iron and steel-producing countries and regions such as Saudi Arabia, the United States, India, and Europe. It will also assess how emerging global trade in green iron, driven by hydrogen-based technologies, could reshape the global steel production landscape and trade flows. The goal is to offer insights for the future transformation of the global iron and steel industry and support the development of green iron trade and investment strategies.

Like other sectors, heavy industries such as iron and steel must rely on a combination of technological efficiency improvements, circular economy practices, and energy substitution to achieve deep decarbonization. The future energy system will be characterized by a dominant share of non-fossil energy sources on the supply side and a high degree of electrification on the demand side. This transformation will heavily depend on critical minerals essential to wind, solar, electric vehicles, and grid infrastructure. Ensuring the sustainable supply of these minerals is vital to meeting climate goals on schedule.

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Copper plays a foundational and strategic role in modern industry and energy systems. Its excellent conductivity, thermal performance, ductility, and corrosion resistance make it indispensable across a wide range of sectors—including power transmission and distribution, transportation, construction, electronics, and household appliances—making it integral to virtually every key industrial value chain. Against the backdrop of global energy transition and climate action, copper demand is set to rise rapidly due to its essential role in green energy infrastructure (e.g., wind, solar, storage, and grids) and end-use applications such as electric vehicles. As the world's largest consumer and producer of refined copper, China will see its copper demand and supply dynamics directly influencing the trajectory of its energy transition and climate targets.

The proposed study will be completed within 24 calendar months. TWO progress reports will be submitted during the duration of the proposed study and the final report will be submitted after the project is completed.

3. OBJECTIVES

This project has two primary objectives:

(1) To develop an optimization analysis framework for hydrogen-based steelmaking technologies that incorporates both regional characteristics and global implications. The framework aims to support decarbonization and low-carbon technology deployment in major steel-producing regions worldwide, while also providing insights for assessing emerging patterns in global steel trade.

(2) To investigate China's copper supply chain—focusing on copper resource flows, material metabolism, and medium- to long-term demand evolution—in order to better understand the supply-demand dynamics and potential risks associated with this critical mineral in the energy transition.

To achieve this objective, the scope of the work are as follows:

3.1 Literature Review: Conduct a comprehensive review of existing research and technical reports on low-carbon steel technologies in key countries and regions, as well as studies on copper resource flows and metabolism in China, to build a foundational understanding of industry characteristics and development patterns.

3.2 Professional Exchange: Organize a series of "Steel +" and "Copper+" salons. These sessions will invite experts and institutions from energy, steel, hydrogen, and copper sectors to provide guidance, insights, and discussion on industry development trends and emerging technologies, helping to consolidate both shared perspectives and regional differences.

3.3 Database Development: Develop spatial-temporal databases on steel technologies and renewable energy systems in key countries or regions (e.g., Saudi Arabia, the United States, India), and compile detailed datasets on copper supply chains and demand flows within China.

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3.4 **Field Research**: Explore the current status, the challenges of emission reduction, and ongoing practices such as DRI of steel companies. Analyze the technological preferences of different regions and enterprises based on their roadmaps.

3.5 **Model Optimization and Extension:** extend the hydrogen-based steelmaking flexibility model developed in Phase I, and conduct comparative techno-economic assessments across target countries and regions. Simultaneously, develop a copper demand forecasting model to evaluate future scenarios for China's copper resource requirements under different carbon neutrality pathways.

4. STATEMENT OF THE WORK

The research will specifically include the following tasks:

4.1 Task 1: Analysis of the Iron and Steel Industry and Renewable Energy Development in Key Countries and Regions

4.1.1 Conduct an in-depth investigation of major steel-producing regions outside China—such as the Middle East (e.g., Saudi Arabia), India, the European Union, and the United States—to assess the current industrial landscape, technological capabilities, and development trajectories of their steel sectors.

4.1.2 Analyze regional differences in renewable energy resource endowments and equipment costs, and build a comprehensive database to support model parameterization.

4.2 Task 2: Modeling of Hydrogen-Based Iron and Steelmaking Systems in Key Countries and Regions

4.2.1 Build upon the Phase I model, which focused on the Chinese market, by refining and expanding it to reflect country- and region-specific production conditions and market environments.

4.2.2 Investigate how major steel-producing countries and regions respond to renewable energy intermittency, optimize the configuration of production processes, simulate regionally customized production models, and calculate optimal production costs.

4.2.3 Compare the techno-economic performance of various steelmaking pathways—particularly hydrogen-based steelmaking—across different countries and regions.

4.3 Task 3: Impact of Global Policy Factors on Iron and Steel Trade

4.3.1 Based on the techno-economic assessments of steelmaking in key countries, analyze the potential influence of policy mechanisms such as U.S. tariffs and the EU's Carbon Border Adjustment Mechanism (CBAM) on global iron and steel trade.

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4.3.2 Examine how cross-national policy differences may reshape global competitiveness and the division of labor within the steel value chain, offering theoretical insights for policymakers.

4.3.3 Explore the emergence of new global trade patterns — especially the shift from large-scale iron ore trade toward hot briquetted iron (HBI) trade.

4.3.4 Conduct case studies of typical global iron ore trade flows (e.g., Australia to China), evaluate the economic viability of alternative trade models, and identify the opportunities and challenges in the future restructuring of the global steel industry.

4.4 Task 4: Copper Demand Forecasting and Supply Risk Analysis in China

4.4.1 Analyze China's current copper resource flows, refining capacity, and external dependency, develop Sankey diagrams for copper metabolism, and identify key supply sources and demand sectors. Compile detailed datasets on copper supply chains and demand flows within China.

4.4.2 Examine copper demand dynamics in emerging sectors such as electric vehicles, energy storage, and smart grids. Predict long-term copper demand trajectories under China's carbon neutrality goals.

4.4.3 Estimate the potential for recycling and recovery of secondary (scrap) copper resources.

4.4.4 Assess future supply-demand imbalances and associated risks by integrating demand forecasts, recycling potential, and import dependencies—offering insights for policy design.

5. DELIVERABLE ITEMS

5.1 Phase 1: July 21, 2025 – November 20, 2026 Duration: 16 months

Planned Technical Progress: Task 1, Task 2, and Task 3.

Milestones and Deliverables:

5.1.1 Research Report: Provide an in-depth analysis of the iron and steel industry in key countries and regions, assess the techno-economic feasibility of renewable energy and hydrogen-based steelmaking technologies, and evaluate the potential impact of international policy factors on global iron and steel trade.

5.1.2 Database: Establish a foundational dataset covering technology efficiency, costs, renewable energy equipment costs, and resource endowments for key countries and regions.

5.1.3 Model: Develop a regionally adaptable optimization model for hydrogen-based steelmaking systems with integrated flexibility and economic assessment capabilities.

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5.1.4 Submit a paper to a reputable journal.

5.2 Phase 2: November 21, 2026 – July 20, 2027 Duration: 8 months

Planned Technical Progress: Task 4.

Milestones and Deliverables:

5.2.1 Research Report: Analyze the evolution of copper supply and demand in China and assess related supply chain risks under the energy transition

5.2.2 Model: Develop a copper demand forecasting model for China aligned with carbon neutrality targets

5.2.3 Submit a paper to a reputable journal with high impact factor

5.2.4 Submit final report for all phases

5.3 Project deliverable items list:

Payment No.	Delivery Schedule	Milestone
1	Upon execution of RPO No. 21	Execution of RPO No. 21 by all parties
2	16 months after RPO execution	Acceptance by Aramco of reports and any other deliverables submitted by Tsinghua University in accordance with Milestones 5.1.1, 5.1.2, 5.1.3 and 5.1.4
3	24 months after RPO execution	Acceptance by Aramco of reports and any other deliverables submitted by Tsinghua University in accordance with Milestones 5.2.1, 5.2.2, 5.2.3 and 5.2.4

6. MILESTONE PAYMENT SCHEDULE

This Research Project shall be completed within a period of 24 months after issuance of a Contract Release Purchase Order.

Payment No.	Milestone	Delivery Schedule	% of Lump Sum	Amount ($)
1	Execution of RPO No. 21 by both parties	Upon execution of RPO No. 21	50%	$200,000.00
2	Acceptance by Aramco of reports and any other deliverables submitted by Tsinghua University in accordance with Milestones 5.1.1, 5.1.2, 5.1.3 and 5.1.4	16 months after RPO execution	25%	$100,000.00

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3	Acceptance by Aramco of reports and any other deliverables submitted by Tsinghua University in accordance with Milestones 5.2.1, 5.2.2, 5.2.3 and 5.2.4	24 months after RPO execution	25%	$100,000.00
Total			**100%**	**$400,000.00**

** No payment will be due until thirty (30) days after completion of respective milestones subject to the acceptance of Aramco.*

** Any payment Amount has included all applicable taxes. In case VAT or related tax is applicable in concerned countries, this should be reported to Aramco prior to submission of any invoice. In case withholding tax is applicable in concerned countries, the amount of tax to be withheld shall be deducted from any Amount payable by Aramco before transferring payment.*

7. CONTACT PERSONS

SAUDI ARAMCO Representative for the purposes of administering this work shall be:
Contact Name: Daniel De Castro Gomez
Telephone: +966 13 8808457
E-Mail : daniel.decastrogomez@aramco.com

Tsinghua University Principal Investigator of this CONTRACT shall be:
Contact Name: Prof. Zheng Li
Telephone: +86 010-62795736
E-Mail : lz-dte@tsinghua.edu.cn

Co-Principal Investigator : Dr. Tianduo Peng
Telephone: +86 010-62780968
+86 13671058362
E-Mail: ptd@mail.tsinghua.edu.cn

The Company Representative must be copied on all reports and electronic communication between Tsinghua University and Aramco. Tsinghua University shall identify Contact Person to be available at all times during performance of WORK to respond to Company Representative.

8. MANAGEMENT PLAN

WORK shall be carried out by Tsinghua University. WORK shall be directed by the Tsinghua University Representative and communicated to the Company Representative as per the deliverables listed above and issues relating to any technical and management problems shall be solved through discussions by both parties.

------------------ **END OF SCOPE OF WORK** ---------------

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